

Controlled Windows Endpoint Telemetry Reconstruction

Sysmon and Splunk Cloud Laboratory Case Study

Prepared by	Environment	Report date
Benjamin Alvarez	WIN11-LAB01	27 September 2026

This case study documents a controlled Windows 11 endpoint telemetry exercise using Sysmon and Splunk Cloud. The analysis reconstructed two deliberately generated process chains, tested whether a reused process ID could explain the results, and resolved the parent identity with Sysmon ProcessGuid evidence. This is a telemetry reconstruction lab, not an investigation that began from an unknown production alert.

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Executive Summary

The objective was to validate endpoint telemetry, ingest it into Splunk Cloud, and reconstruct process lineage from the resulting events. Sysmon recorded process creation with command lines, hashes, integrity level, user context, and process GUIDs. Exact ParentProcessGuid comparison showed that the September 25 Notepad event and September 27 cmd.exe event referenced the same PowerShell process instance, rather than two unrelated processes that happened to reuse PID 8536.

Key Findings

- WIN11-LAB01 ran Windows 11 Pro on an ARM64 VMware virtual machine with UEFI Secure Boot enabled.
- Sysmon v15.22 ARM64 and the Sysmon driver were installed and running.
- The initial Splunk dataset contained 89 events, including 53 Sysmon Event ID 1 process creation records.
- PowerShell process instance {50d62ed7-0720-6ab0-f507-000000000700} (PID 8536) launched Notepad PID 13424, which then launched the packaged Notepad process PID 8768.
- The same PowerShell process instance launched cmd.exe PID 12144 with a command that wrote CaseStudy to C:\Users\Public\marker.txt.
- The shared ParentProcessGuid decodes to an approximate start time of 2026-09-20 16:17:36 UTC, before Sysmon was installed on September 25; this explains why the parent's creation event is absent.
- Two PowerShell temporary policy test file events were unrelated to the controlled marker and were excluded from the primary finding.

Assessment

The process chains are fully consistent with the controlled actions performed during the lab. PowerShell and cmd.exe are legitimate tools that also appear in adversary activity, so the command lines, account, timestamps, resulting artifact, and GUID-based lineage were evaluated together. The identical ParentProcessGuid—not PID 8536 alone—establishes the shared parent process. Within the selected snapshots, no corroborating behavior demonstrated persistence, credential access, data exfiltration, or command and control.

Evidence Confidence

Finding	Confidence	Basis
Telemetry collection operated	High	Running Sysmon service and driver plus recent event records
Notepad lineage reconstructed	High	ProcessGuid and ParentProcessGuid values in both Notepad process-create records
Marker command reconstructed	High	Full cmd.exe command line plus the exact PowerShell ParentProcessGuid
No compromise established	Moderate	Controlled context and no corroborating malicious behavior in the selected snapshots

Laboratory Environment

The endpoint was built as a controlled training system named WIN11-LAB01. A clean baseline snapshot was captured before telemetry configuration so the system could be restored after testing.

Component	Configuration
Guest operating system	Microsoft Windows 11 Pro build 26200
Architecture	ARM64 based PC
Virtualization	VMware Fusion virtual machine
Firmware	UEFI with Secure Boot enabled
Network	NAT through the host connection
Endpoint telemetry	Microsoft Sysmon v15.22 ARM64
Analysis platform	Splunk Cloud Search and Reporting

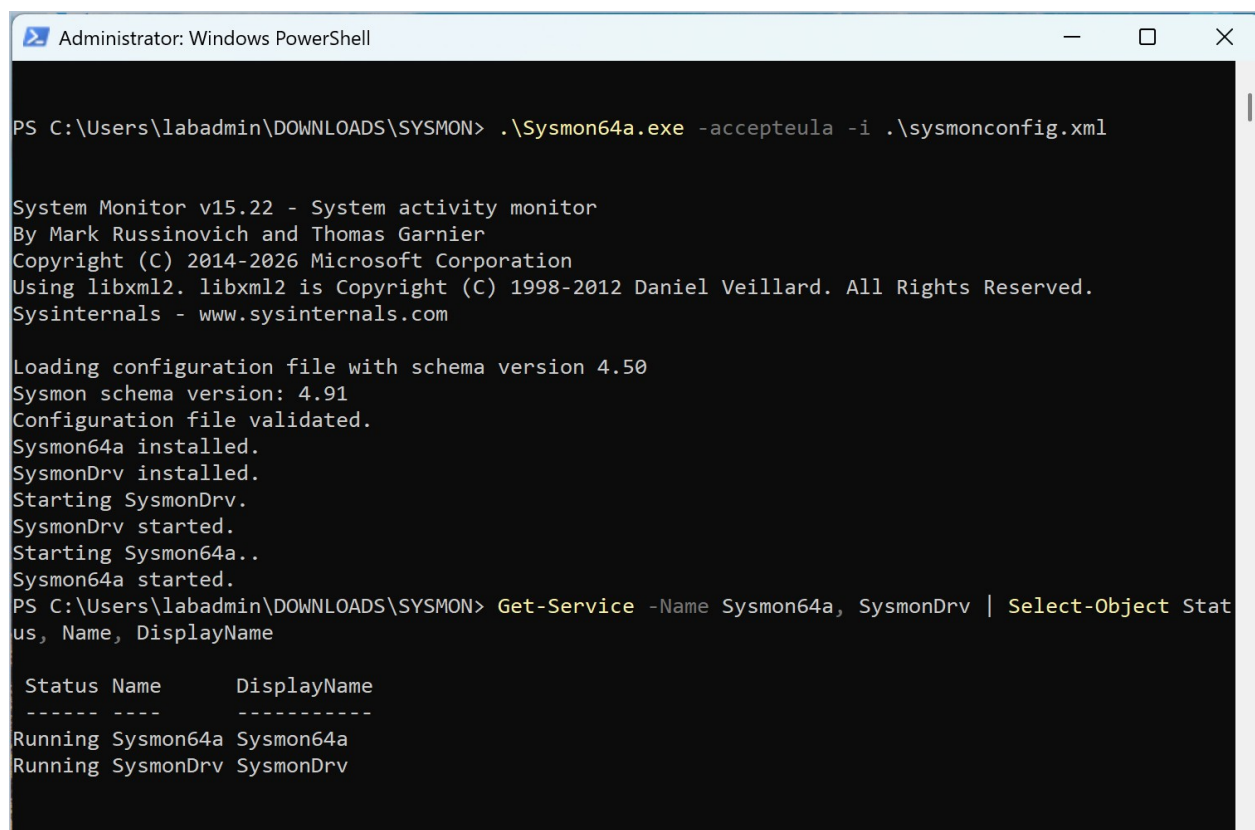
Item	Value
OS Name	Microsoft Windows 11 Pro
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	WIN11-LAB01
System Manufacturer	VMware, Inc.
System Model	VMware20,1
System Type	ARM64-based PC
System SKU	0000000000000001
Processor	Apple silicon, 2000 Mhz, 1 Core(s), 1 Logical Processor(s)
Processor	Apple silicon, 2000 Mhz, 1 Core(s), 1 Logical Processor(s)
Processor	Apple silicon, 2000 Mhz, 1 Core(s), 1 Logical Processor(s)
Processor	Apple silicon, 2000 Mhz, 1 Core(s), 1 Logical Processor(s)
BIOS Version/Date	VMware, Inc. VMW201.00V.25600138.BA64.2607151225, 7/15/2
SMBIOS Version	3.2
Embedded Controller Version	255.255
BIOS Mode	UEFI
BaseBoard Manufacturer	VMware, Inc.
BaseBoard Product	VBSA
BaseBoard Version	1
Platform Role	Desktop
Secure Boot State	On
PCR7 Configuration	Elevation Required to View

Figure 1 WIN11-LAB01 system summary evidence | EV-VM-009

Telemetry Configuration and Validation

Sysmon was installed from the Microsoft Sysinternals ARM64 package on September 25. The configuration loaded successfully, and both the Sysmon64a service and SysmonDrv driver reported a running state. Event validation confirmed that the Microsoft-Windows-Sysmon/Operational log contained recent records, including Event ID 1 process creation events.

The investigation used manual CSV exports that were uploaded to Splunk Cloud. The first export contained 89 events and supported the Notepad lineage exercise. A later export captured the cmd.exe marker test. The embedded Sysmon Message content was treated as the authoritative event narrative because generic CSV serialization also included .NET event object metadata that did not always describe the embedded event cleanly.

A screenshot of a Windows PowerShell window titled "Administrator: Windows PowerShell". The window shows the execution of the Sysmon64a.exe command with the -accepteula and -i flags, followed by a Get-Service command to check the status of Sysmon64a and SysmonDrv. The output shows that both services are installed and running.

```
PS C:\Users\labadmin\DOWNLOADS\SYSMON> .\Sysmon64a.exe -accepteula -i .\sysmonconfig.xml

System Monitor v15.22 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64a installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64a..
Sysmon64a started.
PS C:\Users\labadmin\DOWNLOADS\SYSMON> Get-Service -Name Sysmon64a, SysmonDrv | Select-Object Stat
us, Name, DisplayName

Status Name      DisplayName
-----
Running Sysmon64a Sysmon64a
Running SysmonDrv SysmonDrv
```

Figure 2 Sysmon installation and running service evidence | EV-SYS-004

Investigation Method

The analysis started with a broad host and source search, then narrowed to process creation records and specific command-line terms. Each candidate event was expanded to inspect ProcessGuid, ProcessId, image, command line, user, integrity level, hashes, ParentProcessGuid, ParentProcessId, and ParentImage. The GUIDs were compared across dates specifically to test the alternative explanation of Windows PID reuse.

Step	Action	Purpose
1	Validate the uploaded source and host	Confirm that WIN11-LAB01 data was searchable
2	Filter for Sysmon Event ID 1	Create the process-creation working set
3	Search for Notepad and marker activity	Isolate the two controlled branches
4	Compare ProcessGuid and ParentProcessGuid	Correlate lineage without relying on reusable PIDs
5	Validate the cmd.exe command and artifact	Connect process, command line, and marker-file result
6	Decode the shared GUID time component	Compare the parent start time with Sysmon installation
7	Review nearby PowerShell file events	Exclude routine temporary policy-test files

Process Lineage

Branch	Process or artifact	Relationship	Evidence
Shared parent	powershell.exe PID 8536	Exact ParentProcessGuid match across both dates	EV-INV-003 and EV-INV-005
Notepad	notepad.exe PID 13424	Child of the shared PowerShell instance	EV-INV-003
Notepad	Notepad.exe PID 8768	Child of Notepad PID 13424 by GUID and PID	EV-INV-004
Marker	cmd.exe PID 12144	Child of the shared PowerShell instance	EV-INV-005
Marker	C:\Users\Public\marker.txt	Artifact created by the recorded command	EV-INV-005 and local verification
Identity check	ParentProcessGuid ...0720-6ab0...	Start component resolves to 2026-09-20 16:17:36 UTC	EV-INV-003 and EV-INV-005

Process Identity Resolution

Both branches record ParentProcessGuid {50d62ed7-0720-6ab0-f507-000000000700}. For a reproducibility check, the GUID time fields 3 and 2 form hexadecimal 6ab00720, which converts to approximately 2026-09-20 16:17:36 UTC. The same calculation reproduces the observed seconds for all three child events. Because Sysmon was not installed until September 25, the shared parent's creation event falls outside the available telemetry window.

☒ Field	Value	Actions
☒ host ▾	WIN11-LAB01	▾
☒ source ▾	sysmon_events.csv	▾
☒ sourcetype ▾	csv	▾
☐ Id ▾	1	▾
☐ MachineName ▾	WIN11-LAB01	▾
☐ Message ▾	Process Create: RuleName: - UtcTime: 2026-09-25 16:47:22.996 ProcessGuid: {50d62ed7-a59a-6ab6-410c-000000000700} ProcessId: 13424 Image: C:\Program Files\WindowsApps\Microsoft.Windows Notepad_11.2607.14.0_arm64__8wekyb3d8bbwe\Notepad\Notepad.exe FileVersion: - Description: - Product: - Company: - OriginalFileName: - CommandLine: "C:\WINDOWS\system32\notepad.exe" CurrentDirectory: C:\Users\labadmin\DOWNLOADS\SYSMON\ User: WIN11-LAB01\labadmin LogonGuid: {50d62ed7-3b45-6aaf-877b-3a0000000000} LogonId: 0x3A7B87 TerminalSessionId: 2 IntegrityLevel: High Hashes: MD5=A43E91DC3ACADEF305AE1FED7CCBBD7D,SHA256=229E2D1684E89F054C0907F73028B11819124376A3EB7B87027E996B16747C6,IMPHASH=6BA2C46F1FFB894B4232DBD3E8BAC842 ParentProcessGuid: {50d62ed7-0720-6ab0-f507-000000000700} ParentProcessId: 8536 ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" ParentUser: WIN11-LAB01\labadmin	▾

Figure 3 Notepad PID 13424 and the shared PowerShell ParentProcessGuid | EV-INV-003

☒ Field	Value	Actions
☒ host ▾	WIN11-LAB01	▾
☒ source ▾	sysmon_events.csv	▾
☒ sourcetype ▾	csv	▾
☐ Id ▾	1	▾
☐ MachineName ▾	WIN11-LAB01	▾
☐ Message ▾	Process Create: RuleName: - UtcTime: 2026-09-25 16:47:23.089 ProcessGuid: {50d62ed7-a59b-6ab6-430c-000000000700} ProcessId: 8768 Image: C:\Program Files\WindowsApps\Microsoft.Windows Notepad_11.2607.14.0_arm64__8wekyb3d8bbwe\Notepad\Notepad.exe FileVersion: - Description: - Product: - Company: - OriginalFileName: - CommandLine: "C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2607.14.0_arm64__8wekyb3d8bbwe\Notepad\Notepad.exe" /SESSION:mtTJ3faHOU2MYcmFnm6xxQAAAAABAgABcDQAAPaHOU08AwAAAAA== CurrentDirectory: C:\Users\labadmin\DOWNLOADS\SYSMON\ User: WIN11-LAB01\labadmin LogonGuid: {50d62ed7-3b45-6aaf-877b-3a0000000000} LogonId: 0x3A7B87 TerminalSessionId: 2 IntegrityLevel: High Hashes: MD5=A43E91DC3ACADEF305AE1FED7CCBBD7D,SHA256=229E2D1684E89F054C0907F73028B11819124376A3EB7B87027E996B16747C6,IMPHASH=6BA2C46F1FFB894B4232DBD3E8BAC842 ParentProcessGuid: {50d62ed7-a59a-6ab6-410c-000000000700} ParentProcessId: 13424 ParentImage: C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2607.14.0_arm64__8wekyb3d8bbwe\Notepad\Notepad.exe ParentCommandLine: "C:\WINDOWS\system32\notepad.exe" ParentUser: WIN11-LAB01\labadmin	▾
☐ ProviderName ▾	Microsoft-Windows-Sysmon	▾
☐ TimeCreated ▾	9/25/2026 9:47:23 AM	▾

Figure 4 Packaged Notepad PID 8768 linked to Notepad PID 13424 | EV-INV-004

Investigation Timeline

Times below use UTC when the Sysmon UtcTime value was available. The Windows interface displayed Pacific daylight time, seven hours behind UTC on the investigation dates.

Time	Event	Finding	Evidence
2026-09-25 16:47:22.996 UTC	Process Create	Shared PowerShell instance launched Notepad PID 13424	EV-INV-003
2026-09-25 16:47:23.089 UTC	Process Create	Notepad PID 13424 launched packaged Notepad PID 8768	EV-INV-004
2026-09-27 14:46:08.129 UTC	Process Create	Shared PowerShell instance launched cmd.exe PID 12144	EV-INV-005
2026-09-27 14:46:08.129 UTC	Command line	cmd.exe wrote CaseStudy to C:\Users\Public\marker.txt	EV-INV-005

Decisive Evidence

The Notepad and cmd.exe Event ID 1 records contain the identical PowerShell ParentProcessGuid {50d62ed7-0720-6ab0-f507-00000000700}. That exact identifier rules out ordinary PID reuse as the reason both events list PID 8536. The cmd.exe event also records the complete marker command, executing user, high integrity, hashes, and parent image; local verification returned the expected CaseStudy text.

```
: 9/27/2026 7:46:08 AM
: Process Create:
  RuleName: -
  UtcTime: 2026-09-27 14:46:08.129
  ProcessGuid: {50d62ed7-2c30-6ab9-8910-00000000700}
  ProcessId: 12144
  Image: C:\Windows\System32\cmd.exe
  FileVersion: 10.0.26100.9278 (WinBuild.160101.0800)
  Description: Windows Command Processor
  Product: Microsoft® Windows® Operating System
  Company: Microsoft Corporation
  OriginalFileName: Cmd.Exe
  CommandLine: "C:\WINDOWS\system32\cmd.exe" /c "echo
CaseStudy>C:\Users\Public\marker.txt"
  CurrentDirectory: C:\Users\labadmin\DOWNLOADS\SYSMON\
  User: WIN11-LAB01\labadmin
  LogonGuid: {50d62ed7-3b45-6aaf-877b-3a0000000000}
  LogonId: 0x3A7B87
  TerminalSessionId: 2
  IntegrityLevel: High
  Hashes: MD5=9533CA888121A14082B314674F27EBAE,SHA256=DC2C9B12465AC5833BDC418B66C7C2C
71365AA548BD9BA641EF3C5A0F3DB4D81,IMPHASH=F408133B959AA5692E76A0DA7CD221F8
  ParentProcessGuid: {50d62ed7-0720-6ab0-f507-00000000700}
  ParentProcessId: 8536
  ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
  ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
  ParentUser: WIN11-LAB01\labadmin
```

Figure 5 cmd.exe marker command and shared PowerShell parent | EV-INV-005

Technical Assessment

Supported Conclusion

The selected activity was produced by the analyst during controlled telemetry validation. The command lines, timestamps, account, process GUIDs, parent-child relationships, and marker contents agree with the documented test steps. The evidence supports one shared, long-lived PowerShell parent and a benign lab classification for the selected events.

Alternative Explanations

Windows can reuse process IDs, so PID 8536 alone could not prove that events two days apart shared a parent. That alternative was tested and ruled out because both child events record the exact same ParentProcessGuid and PowerShell image. The GUID's time component predates Sysmon installation, explaining the absent parent-creation record. The evidence does not establish why the PowerShell process remained active. Separate __PSScriptPolicyTest temporary files were routine policy checks and did not match the marker path.

MITRE ATT&CK Context

Technique	Observed behavior	Interpretation
T1059.001 PowerShell	PowerShell served as the parent shell for controlled process launches	Technique relevant for detection engineering but benign in this test
T1059.003 Windows Command Shell	cmd.exe executed an echo command that created marker.txt	Command shell use was deliberate and locally verified

Limitations

- The dataset consisted of manual CSV snapshots rather than continuous endpoint forwarding.
- The parent PowerShell process began before Sysmon installation, so its Event ID 1 creation record could not have been collected. The evidence confirms the shared parent identity but not why that process remained active.
- No memory image, full disk image, packet capture, or enterprise identity telemetry was collected.
- The conclusion applies to the selected events and should not be generalized beyond the available evidence.

Recommendations

- For the next alert-driven case, forward Sysmon continuously with the Splunk Universal Forwarder or another supported collector.
- Use parsed EventCode, ParentImage, Image, and CommandLine fields, then build reusable field-based SPL searches and a Sigma rule rather than relying on raw-message terms.
- Generate controlled attacker-like behavior with Atomic Red Team, begin from an alert, and close the case with a SOC-style disposition and recommended action.
- Preserve ProcessGuid and ParentProcessGuid values, normalize timestamps to UTC, and retain the endpoint timezone in case notes.

Evidence Index

The following evidence items support the conclusions in this report. Original screenshots and SHA256 values are maintained in the case evidence register.

Evidence	Description	Use in report
EV-VM-009	WIN11-LAB01 Windows system summary	Laboratory environment
EV-SYS-003	Sysmon installed successfully on 2026-09-25	Parent-process telemetry-window comparison
EV-SYS-004	Sysmon64a service and SysmonDrv driver running	Telemetry configuration
EV-TEL-001	Recent Sysmon events displayed in PowerShell	Telemetry validation
EV-TEL-006	Initial Splunk search returned 89 events	Ingestion validation
EV-INV-001	Fifty three process creation events isolated	Investigation baseline
EV-INV-003	Notepad PID 13424 with shared PowerShell ParentProcessGuid	Notepad lineage and parent identity
EV-INV-004	Packaged Notepad PID 8768 with parent PID 13424	Notepad lineage
EV-INV-005	cmd.exe PID 12144 with marker command and the same ParentProcessGuid	Marker lineage and parent identity

Final Disposition

Disposition: Benign controlled laboratory activity. This foundational case demonstrates telemetry validation, Splunk ingestion, targeted searching, GUID-based process-lineage reconstruction, alternative-explanation testing, evidence handling, and defensible scoping. It does not claim to be an unknown-alert or production incident response case.